

When Trade Drives Markup Divergence: An Application to Auto Markets

Agnes Norris Keiller* Tim Obermeier† Andreas Teichgraeber*
John Van Reenen*

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Abstract

When firms sell in multiple markets, demand-side markup estimates from a single market need not align with supply-side estimates derived from production data that aggregate across markets. Empirical analysis of this divergence has been constrained by a lack of market-specific cost data. We address this by showing that production markups can be expressed as a revenue-weighted average of demand markups across markets and products, which allows the relationship between the two measures to be quantified using revenue shares and markups in domestic and foreign markets—information that is generally more accessible than cost data. Applying this approach to UK car manufacturers, we find that supply-side markups increased between 1998 and 2018, while demand-side markups declined. These trends can be reconciled by an increase in the markup that UK-based producers gained on their exports, which we corroborate using administrative trade data. We find evidence that increases in supply-side markups have been driven by exports, particularly to China where prices are high relative to other export markets. Our results highlight how international market exposure can drive aggregate markups and generate differences between supply- and demand-based measures of market power.

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Data Disclaimer: Parts of this work were produced using statistical data from the UK Office for National Statistics (“ONS”). The use of ONS data does not imply the endorsement of the ONS in relation to its interpretation or analysis. Analysis using ONS research datasets may not exactly reproduce ONS aggregates and was carried out in the Secure Research Service, part of the Office for National Statistics.

*Programme on Innovation and Diffusion, London School of Economics

†University of Leicester

1 Introduction

Product market power has long been of interest to economists and policy makers. While the difference between a firm’s price and marginal cost — the markup — is a convenient summary measure of market power (Berry et al. 2019), it is rarely observed directly due to a lack of marginal cost data. As a result, several estimators have been proposed that recover markups by exploiting the first-order conditions of profit maximization. These methods can be broadly grouped into two approaches. The first, which we label the “demand” approach uses data on prices, purchase decisions, and consumer attributes to estimate price elasticities of demand. These, coupled with assumptions on firms’ pricing behaviour, identify marginal costs and hence markups (Berry et al. 1995, 2004; Grieco et al. 2024). The second “production” or supply-side approach uses data to estimate production function parameters which identify markups from the first-order condition of a variable input choice (Hall 1988; De Loecker and Warzynski 2012). These approaches have distinct advantages and disadvantages, which are well-discussed in the literature (see below) as they require distinct assumptions and data. A prominent recent literature finds increases in aggregate markups when measured from the production side (De Loecker et al. 2020), but not so clearly from demand-side estimation (e.g., Grieco et al. (2024)) and this has led to dispute over methodological and data issues (e.g., Benkard et al. (2025) and De Loecker et al. (2025)).

In this paper, we emphasize that demand and production markup estimates can diverge when firms sell to multiple markets with heterogeneous preferences, *even when the identifying assumptions of both estimators hold*. This divergence is due to differences in the data each approach uses. While demand-based estimation typically uses data on purchases and prices consumed within a particular country, production-based estimation usually uses data relating to the output of establishments located in a single country even if this output is exported to overseas markets. Given that firms, even within narrowly-defined industries, often sell multiple products into multiple national markets, it is clear that markups estimated by either approach will not necessarily align.

To make this explicit, we show that production markups - defined either as the price-to-marginal cost ratio or the Lerner index - can be expressed as revenue-weighted averages across markets (and products). De Loecker and Warzynski (2012) show a similar aggregation. They express producer-side markups as a *cost share*-weighted average of market-specific markups in order to provide suggestive evidence that firms obtain heterogeneous markups when they sell in foreign rather than domestic markets. However, our alternative aggregation offers the practical advantage that, unlike cost-shares, market-specific revenue shares are often observed either in firms’ financial reports or in trade data. This allows a decomposition of the total production-based markup into a domestic and a foreign component.

To demonstrate the utility of our method, we use data from the UK car market to estimate markups using both demand and production approaches. This is an attractive application for several reasons. Cars are commonly the second most expensive purchase for consumers (after housing), and hence markups can have substantial impacts on con-

sumer welfare. Second, many papers on demand-based markup estimation focus on the car market, providing a benchmark for our findings. Third, other work shows that UK car manufacturers exhibited relatively strong productivity growth over the last four decades, which translated into higher wages (Norris Keiller et al. 2024). The interpretation of this performance as a ‘success story’ would be tempered if it were due to increased markups rather than real productivity gains.

We compile a comprehensive data set to estimate both demand and production markups for the period 1998 to 2018. On the production side, we use data from Historical Orbis (HO), which provides firm-level financial data. For the demand side, we utilize product-level information that includes prices, quantities sold, and other product characteristics from the market intelligence firm JATO. Applying both methodologies, we observe a divergence: the production-side Lerner index rises over time, while the demand-side Lerner index declines.¹

We then show that these divergent trends can be reconciled by an increase in the markups earned by UK-based car manufacturers on their export sales. About 80% of cars produced in the UK were exported, making the distinction between domestic and foreign markets particularly relevant. If UK-based plants earn higher markups because a substantial share of output is sold in high-markup destination markets, then declining domestic markups can coexist with rising overall markups across all markets. We quantify the required increase in the rest-of-world markup needed to reconcile our supply and demand estimates. To address the possibility that the divergence reflects other factors, such as violations of the market conduct assumptions underlying demand-side estimation, we validate our interpretation using customs data on exports. We show that increases in firm-level markups are positively associated with growth in the revenue shares of exports to China and the United States; destinations that expanded rapidly over our sample period. This evidence is consistent with the interpretation that an increasing share of exports to destination markets characterized by relatively high markups—arising, for example, from more price-inelastic demand for certain UK-produced vehicles—has contributed to higher production markups. Overall, our findings suggest that rising markups earned by UK-based car manufacturers have not come at the expense of UK consumers.

Related literature

This paper makes both methodological and empirical contributions to the literature on markup measurement and market power. Methodologically, we link production and demand markup estimates through a simple aggregation formula. This framework connects the firm-level production-side approach developed by De Loecker and Warzynski (2012), building on the industry-level insights of Hall (1988), with the demand-side estimates pioneered by Berry et al. (1995). By clarifying how these estimators relate to one another in the presence of multiple markets, our approach provides a unified perspective on two

1. Declines in demand-based markups have also been found in the car market in the US (Grieco et al. 2024).

widely used methods for measuring markups.

Empirically, we document trends in production-based and demand-based markups in the UK car market over the period 1998–2018. Consistent with the economy-wide evidence in De Loecker et al. (2020) and De Loecker et al. (2022), we find that production-based markups among UK car manufacturers increased over this period. In contrast, demand-based markups declined, generating a divergence similar to that documented by Grieco et al. (2024) for the US car market. While Grieco et al. (2024) interpret this divergence as evidence of shortcomings in production-based methods, we propose an alternative explanation that emphasizes the role of export markets in a smaller, more open economy such as the UK.²

Whereas much of the existing literature analyzes production-based or demand-based markups in isolation, our work is also closely related to De Loecker and Scott (2022), who compare markups obtained from both approaches using data from the US brewing industry. Their analysis highlights how vertical structure can generate differences between producer- and demand-based markups when retailers face a different competitive environment than manufacturers. In contrast, our paper focuses on international trade as a source of divergence. While De Loecker and Scott (2022) find rising markups under both approaches and interpret this as evidence consistent with highly competitive retail markets, we show that production-based and demand-based markups may diverge even in the presence of competitive downstream sectors, if a substantial share of production is exported to markets with different demand conditions.³

Our paper also contributes to the literature on pricing, product differentiation, and international market segmentation in the automobile industry, particularly in relation to market power (e.g. Goldberg 1995; Verboven 1996; Goldberg and Verboven 2001). Related work by Berry et al. (1999) and Lacetera and Sydnor (2015) further emphasizes the importance of production location and trade patterns in shaping outcomes in the car market.

The remainder of the paper proceeds as follows. Section 2 presents the aggregation framework, showing that production markups can be expressed as a revenue-weighted average across products and destination markets, and derives a dynamic decomposition that separates changes in markups into within- and between-market components. Section 3 describes the empirical application. We provide background on the UK car market, detail the data and methods used to estimate production and demand markups, and explain how the aggregation framework is used to infer export markups. We then present our main results, alongside complementary evidence from administrative trade data that corroborates the increase in markups on exports implied by our aggregation formula. Section 4 concludes.

2. Recent work examining trends in demand-side markups in other industries includes studies of cement (Miller et al. 2023) and consumer packaged goods (Atalay et al. 2023; Döpper et al. 2025).

3. Accordingly, the test of retailer competition proposed by De Loecker and Scott (2022) is best suited to industries in which the vast majority of output is sold domestically. Input–output tables indicate that approximately 98% of US-produced beer is consumed domestically, rendering trade considerations second-order in their setting.

2 Reconciling demand and production markup estimates

We start from the commonly-used assumption that markets can be defined as a country-product combination. Following from this, the price sensitivity of consumers in any market in combination with assumptions on firms' price-setting behaviour can be used to identify demand markups. Production markup estimation, by contrast, relies on firm-level data and since firms often sell multiple products into multiple national markets, estimates obtained via the two approaches are likely to diverge. In this section, we make this explicit by showing how firm-level (i.e. production) markups can be expressed as revenue-weighted averages of county-product specific demand markups. The exact expression of this aggregation depends on whether one defines markups using the Lerner index (i.e. the difference between price and marginal cost divided by price), or the ratio of price to marginal cost (PMC). We focus on Lerner index markups in our main analysis, since the aggregation formula is more straightforward, but show how to decompose PMC markups in Appendix A.

Suppose firm i serves J perfectly separable national markets, $j \in [1, \dots, J]$. For each product, $k \in [1, \dots, K]$, the Lerner-index markup obtained by firm i in market j is

$$\mu_{ijk}^L = \frac{p_{ijk} - c_{ijk}}{p_{ijk}}, \quad (1)$$

where c denotes marginal cost and p denotes price. Multiplying the right hand side of equation (1) by $\frac{q_{ijk}}{q_{ijk}} = 1$ gives

$$\begin{aligned} \mu_{ijk}^L &= \left(\frac{p_{ijk} - c_{ijk}}{p_{ijk}} \right) \left(\frac{q_{ijk}}{q_{ijk}} \right) \\ &= \frac{R_{ijk} - C_{ijk}}{R_{ijk}}, \end{aligned} \quad (2)$$

where R denotes revenue and C denotes cost of goods sold.⁴

Define firm i 's market-specific aggregate markup in market j as

$$\begin{aligned} \mu_{ij}^L &= \frac{R_{ij} - C_{ij}}{R_{ij}} = \sum_{k=1}^K \frac{R_{ijk} - C_{ijk}}{R_{ij}} = \sum_{k=1}^K \left(\frac{R_{ijk}}{R_{ij}} \right) \left(\frac{R_{ijk} - C_{ijk}}{R_{ijk}} \right) \\ &= \sum_{k=1}^K \left(\frac{R_{ijk}}{R_{ij}} \right) \mu_{ijk}^L. \end{aligned} \quad (3)$$

This shows that the aggregate markup which firm i obtains from market j is the revenue-weighted average of product-specific markups. Using a similar logic, we can aggregate

4. For simplicity, this derivation assumes constant marginal costs with respect to quantity ($MC(q) = c$). In Appendix B, we show that a similar formula can be derived without making assumptions on the shape of the marginal cost curve.

over markets to obtain firm i 's overall aggregate markup as a revenue-weighted average of their market-specific aggregate markups

$$\begin{aligned}\mu_i^L &= \frac{R_i - C_i}{R_i} = \sum_{j=1}^J \frac{R_{ij} - C_{ij}}{R_i} = \sum_{j=1}^J \left(\frac{R_{ij}}{R_i} \right) \left(\frac{R_{ij} - C_{ij}}{R_{ij}} \right) \\ &= \sum_{j=1}^J \left(\frac{R_{ij}}{R_i} \right) \mu_{ij}^L.\end{aligned}\tag{4}$$

This shows that the conceptual difference between demand and production markup estimates: the data deployed by the former estimate a market-product-specific μ_{ijk}^L , whereas the latter seek to estimate the aggregate firm-specific μ_i^L .

As is the case with other aggregate concepts, one can attribute changes in firm-level aggregate markups to changes in market revenue shares ("across-market" changes) and changes in the market-specific markups ("within-market" changes). Explicitly, defining $r_{ij} = \frac{R_{ij}}{R_i}$ for parsimony, firm i 's aggregate markup can be expressed as

$$\begin{aligned}\Delta\mu_{it}^L &= \mu_{it}^L - \mu_{it-1}^L \\ &= \sum_j r_{ijt} \mu_{ijt}^L - \sum_j r_{ijt-1} \mu_{ijt-1}^L \\ &= \sum_j \underbrace{r_{ijt} (\mu_{ijt}^L - \mu_{ijt-1}^L)}_{\text{due to within-market changes in } j} \\ &\quad + \sum_j \underbrace{(r_{ijt} - r_{ijt-1}) \mu_{ijt-1}^L}_{\text{due to reallocation toward market } j},\end{aligned}\tag{5}$$

where the final equality follows from adding and subtracting $\sum_j r_{ijt} \mu_{ijt-1}^L$.

3 Empirical application: the UK car market

3.1 Historical overview of the UK car market

The supply- and the demand-side of the UK car market have undergone significant changes over recent decades. Car manufacturing in the UK went through significant challenges during the 1970s including labour strikes, financial difficulties, and increasing competition from international manufacturers. British Leyland, formed through the merger of several companies, struggled with inefficiencies and quality issues and was eventually nationalised. More positively, the 1980s marked the beginning of foreign investment in the UK, with Japanese manufacturers such as Nissan and Toyota establishing production facilities to benefit from the UK's skilled workforce and strategic location within Europe. Foreign investment and ownership continued through the 1990s as Rover, formerly part of British Leyland, was sold to BMW, while American-owned Ford acquired the luxury

brands Jaguar and Land Rover. The presence of Japanese manufacturers grew, with Honda joining Nissan and Toyota in setting up significant operations.

By the mid-1990s, when our production data begins, Ford accounted for approximately 25% of revenue in the industry with MG Rover representing just over 20% and Vauxhall 15%. Figure 1a shows that MG Rover’s market share fell sharply in the late 1990s and early 2000s as BMW sold off constituent brands retaining only Mini, which it revitalised under its own branding. Japanese manufacturers Toyota and Nissan increased their share of industry revenue during the early-mid 2000s but the largest change in the industry occurred in 2008 when Tata Motors acquired Land Rover and Jaguar. The newly-branded Jaguar-Land Rover (JLR) increased production dramatically over the following years, primarily via expansion into overseas markets. Land Rover became financially integrated into JLR in 2014 and by 2015 the combined company accounted for approximately 37% of industry revenue.

Trends in car purchases have also undergone marked changes in recent decades as preferences have shifted away from Western brands and technological changes have led to the introduction of different models. The 1980s and 1990s were characterised by a decline in the share of UK registrations accounted for by British and American brands, with Japanese firms such as Nissan, Toyota and Honda rising in popularity as consumers favoured compact and mid-size vehicles. In the late 1990s, when our consumer data starts, Ford was the dominant brand, accounting for 20% of registrations with Vauxhall and Nissan each accounting for 13%. Demand shifted toward luxury and performance vehicles in the early 2000s benefiting brands such as BMW and Audi, whose market shares rose from 2-3% in 2000 to around 5% in 2007. The financial crisis of 2008 affected sales briefly, but the market quickly rebounded. Ford and Vauxhall’s popularity waned in subsequent years as consumers started to favour larger SUV models and South Korean brands, such as Kia.

Given the marked changes in composition, both on the supply- and demand-side of the UK car market shown in Figure 1, it is plausible that markups in the industry have also changed. We now examine this directly.

3.2 Markups in the UK car market

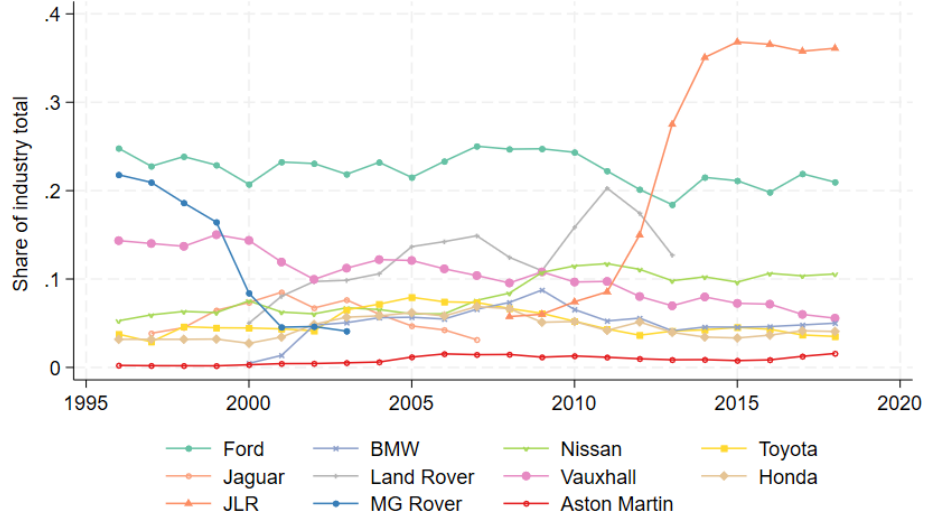
Consider a two-market application of equation (4), for a UK-based manufacturer which distinguishes the domestic (UK) market from the rest of the world (ROW). This gives:

$$\mu_i^L = \left(\frac{R_{iUK}}{R_i} \right) \mu_{iUK}^L + \left(\frac{R_{iROW}}{R_i} \right) \mu_{iROW}^L. \quad (6)$$

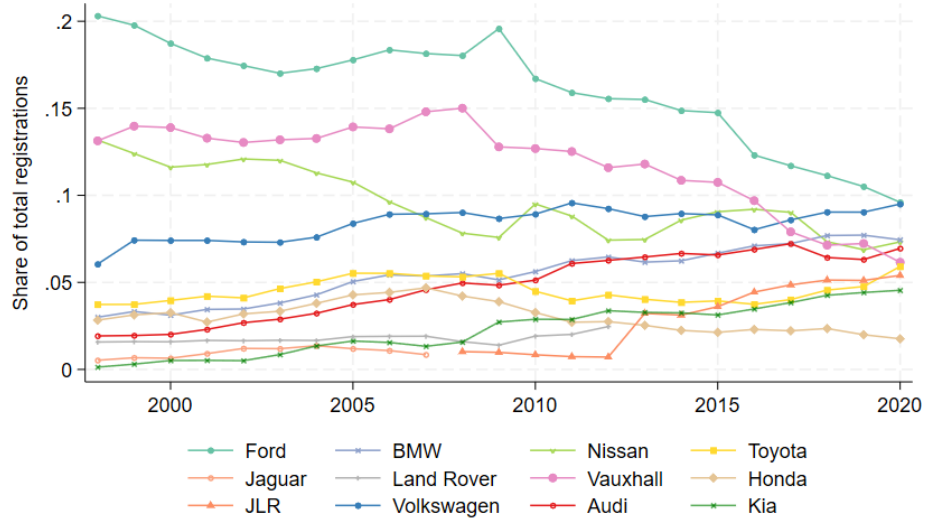
We use data from the UK car industry to demonstrate the utility of this simplified markup aggregation formula as a method of reconciling divergent trends in production and demand markup estimates. This subsection describes the data and methods used to calculate each component of equation (6).

Figure 1: UK Car Market Shares

(a) Supply-side Revenue Shares



(b) Demand-side Registration Shares



Note: Figure 1a shows car manufacturing industry revenue shares of major brands. Figure 1b shows the share of all new annual car registrations accounted for by major brands. Series names pertain to distinct companies and do not reflect ultimate ownership.

3.2.1 Production markups

To obtain production markup estimates $\hat{\mu}_i^L$, we first estimate PMC markups using the methodology of De Loecker and Warzynski (2012) and use it to infer the Lerner index as $\hat{\mu}_i^L = 1 - (1/\hat{\mu}_i^{PMC})$. De Loecker and Warzynski (2012) extend Hall (1988) to show that the first-order condition of a firm's cost minimisation problem taken with respect to a flexible input can be rearranged to obtain

$$\mu_i^{PMC} = \theta_i^X \frac{R_i}{C_i}.$$

We use Historical Orbis (HO) data on UK-based car manufacturers to implement this method.⁵ HO data is derived from firms' financial documents and contains measures of input costs and revenue. We provide further details on data, including sample summary statistics, in Appendix C.

We take materials (defined as cost of goods sold minus the wagebill) to be our flexible input and assume the production technology is Cobb-Douglas as well as constant across firms and time. These assumptions impose $\theta_i^X = \theta^X$, which we estimate using variants on the log-linearised gross output production function:

$$y_{it} = \theta^0 + \theta^M m_{it} + \theta^W w_{it} + \theta^K k_{it} + \tau_t + \epsilon_{it}, \quad (7)$$

where y , m , w and k denote the log of revenue, materials, wagebill and fixed assets respectively. τ is a full set of year dummies and ϵ is an unobserved productivity shock.

Table 1: Production Function Estimates

	(1)	(2)	(3)	(4)	(5)
	OLS Levels	OLS 1st diffs	OLS FE Levels	OLS Levels	OP
Ln(Materials)	0.893*** (0.052)	0.919*** (0.076)	0.899*** (0.062)	0.891*** (0.053)	0.915*** (0.046)
Ln(Wagebill)	0.066 (0.052)	0.092* (0.047)	0.096** (0.041)	0.061 (0.050)	0.038*** (0.013)
Ln(Fixed Assets)	0.072 (0.046)	-0.029 (0.029)	0.075* (0.038)	0.085 (0.048)	0.054*** (0.006)
Year FEs	Yes	Yes	Yes	Yes	Yes
Firm FEs	No	No	Yes	No	No
N obs.	169	158	169	160	160
N firms	10	10	10	10	10

Note: standard errors in parentheses clustered at the firm level. */**/** denote significance at the 10/5/1 percent level respectively.

5. Analysis is restricted to firms observed in the data used to estimate demand markups described in subsection 3.2.2. Although a subset of all UK-based car manufacturers, figure C.1 in Appendix C.1 show these firms account for between 81 and 88 % of total annual revenue in the industry over our analysis period.

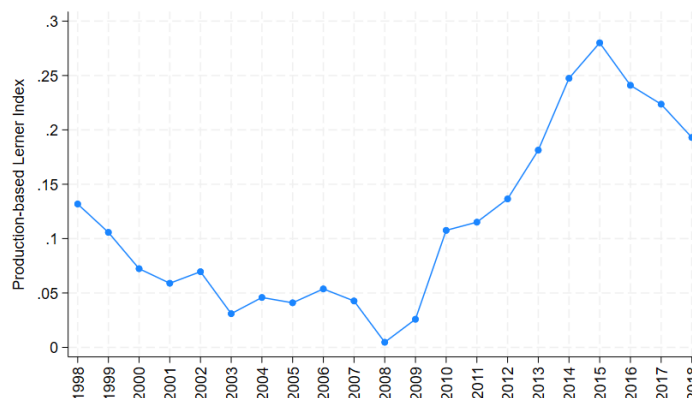
Table 1 contains estimates of the input parameters of equation (7) obtained from various estimators. Columns (1)-(3) contain OLS estimates in levels (column (1)), first-differences (column (2)), and including firm dummies (column (3)). Column (4) repeats column (1) on the sub-sample for which we observe investment, which is needed to implement the Olley-Pakes (OP) control function estimator (Olley and Pakes 1996) of column (5). While the labour and capital coefficients vary somewhat across columns, the materials coefficient is relatively stable and approximately equal to 0.9.⁶ We therefore take 0.9 as our estimate of θ^M and estimate production markup as

$$\hat{\mu}_i^L = 1 - \frac{C_i}{0.9R_i},$$

where C is given as the cost of material purchases observed in the Historical Orbis data.

Figure 2 plots the revenue-weighted mean production markup over the period 1998–2018.⁷ Production markups declined from the late 1990s until the global financial crisis in 2009, after which they rebounded and continued to rise, reaching a peak in 2015, shortly before the Brexit referendum. Although markups fell modestly in the subsequent years, they remained elevated at approximately 20% in 2018—roughly twice their level two decades earlier.

Figure 2: Aggregate Production Lerner index



Note: figure shows the revenue-weighted mean production-based Lerner index estimate. Series are restricted to car brands with manufacturing operations in the UK.

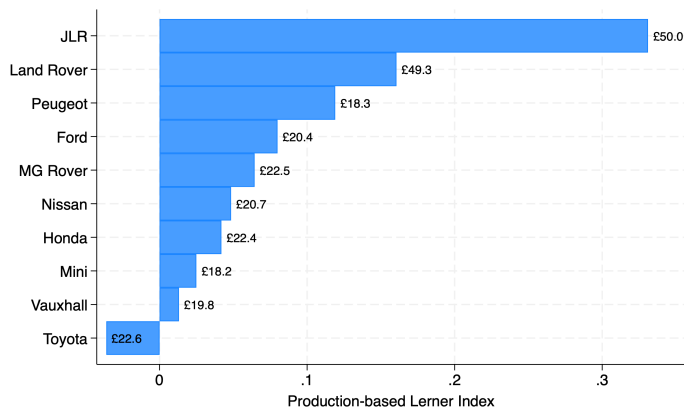
Figure 3 plots production markups by manufacturer, averaging across years. Jaguar Land Rover (JLR) exhibits particularly high markups, consistent with the relatively high average prices of its models (reported in the bar labels). Land Rover’s elevated markups

6. Tables D.1-D.3 in appendix D summarise equivalent results obtained using dynamic panel estimation methods. In all cases, the coefficient on $\log(\text{Materials})$ is approximately equal to 0.9.

7. To parallel our within-firm aggregation formula, the mean markup is calculated using revenue weights.

are similarly unsurprising given its position in the premium segment. In contrast, Peugeot displays the third-highest markup despite having the second-lowest average model price among the manufacturers considered. This implies they are relatively productive and, indeed, the firm fixed-effect estimated via the OLS regression summarised in column (3) of Table 1 is the highest among the manufacturers we consider.

Figure 3: Aggregate Production Lerner index By Manufacturer



Note: figure shows the revenue-weighted mean production-based Lerner index estimate. Bar labels show the mean manufacturer-suggested retail price for models manufactured in the UK observed in the JATO data described in subsection 3.2.2. Prices are expressed in thousands of 2018 pounds and rounded to the nearest £100.

3.2.2 Demand markups

We estimate demand-based markups for the UK using a range of alternative demand system specifications. All estimators require data on product characteristics—most importantly prices—as well as sales quantities. We obtain these data from JATO, a market intelligence company that compiles information on vehicle registrations, manufacturer-suggested retail prices, and other product attributes at the manufacturer–model–country-of-origin level. Our JATO dataset covers all major manufacturers and accounts for approximately 95% of UK vehicle registrations over the period 1998–2022. Additional details on data sources and data construction are provided in Appendix C.

Demand model

We specify a demand system that closely follows the empirical demand literature (Berry 1994; Berry et al. 1995). Our baseline specification employs a nested logit model.⁸ In each year t , households i choose whether to purchase one of K car models, indexed by

8. We also estimate a random coefficients logit model following Berry et al. (1995). The estimated random coefficients are statistically insignificant, yielding results that are quantitatively similar to those obtained from the nested logit specification. Further details are provided in Appendix E.

$k \in 1, \dots, K$, or to select the outside option, denoted by $k = 0$. Models are grouped into nests $b \in 1, \dots, B$, such that models within the same nest are closer substitutes in utility space, conditional on observed characteristics. Households maximise their conditional indirect utility U_{ikt} :

$$U_{ikt} = \beta x_{kt} + \alpha p_{kt} + \xi_{kt} + (1 - \lambda)\zeta_{tb} + \lambda\epsilon_{ikt}, \quad (8)$$

where x_{kt} denotes a vector of characteristics of model k in year t , p_{kt} is the price of model k in year t , ξ_{kt} is an unobserved demand shifter, ζ_{tb} is an unobserved component that is common across all alternatives in the same nest $b \in B$, and ϵ_{ikt} is a demand shock following a Type-1 extreme value distribution (i.i.d. across products).

Equation (8) yields a system of equations that we can solve analytically. Following Berry (1994) and exploiting the properties of the logit, we estimate:

$$\log(s_{kt}) - \log(s_{0t}) = \beta x_{kt} + \alpha p_{kt} + (1 - \lambda) \log(s_{kt|b}), \quad (9)$$

where s_{kt} denotes the market share of product k in year t , s_{0t} is the share of the outside option, and $s_{kt|b}$ is the share of product k within nest b .

Intuitively, the correlation parameter λ measures the degree of substitutability within nests. We require $\lambda \in (0, 1]$ to be consistent with utility maximisation. Note that if the correlation parameter $\lambda = 1$, we are back to the multinomial logit model that does not account for correlation of tastes for unobserved characteristics within a nest. We define markets as respective years t and the market share of product k in market t as the number of sales divided by the number of UK households. The outside option s_{0t} (i.e. not buying a new car) accounts for the largest market share in all years of our sample (around 0.9).

From equation (9), we can derive expressions for own- and cross-price derivatives for each product in each market. These measures of price sensitivity, together with data on ownership, prices, and market shares, form the key inputs for computing product-level markups. Importantly, price derivatives differ depending on whether they correspond to (i) the own-price derivative of product k , (ii) the cross-price derivative of a product k with respect to another product r in the same nest, or (iii) the cross-price derivative of a product k with respect to a product r in a different nest. These derivatives are given by

$$\frac{\partial s_{kt}}{\partial p_{rt}} = \begin{cases} \alpha s_{kt} \left(\frac{1}{\lambda} - \frac{1-\lambda}{\lambda} s_{kt|b} - s_{kt} \right) & \text{if } r = k \\ -\alpha s_{rt} \left(\frac{1-\lambda}{\lambda} s_{kt|b} + s_{kt} \right) & \text{if } r \neq k, \text{ but } r, k \text{ in same nest} \\ -\alpha s_{rt} s_{kt} & \text{otherwise.} \end{cases} \quad (10)$$

Elasticities are calculated using observed prices, market shares, and parameter estimates of the price coefficient α and correlation coefficient λ .

To account for unobserved characteristics that are fixed across time or models, all specifications include model and market (year) fixed effects. Formally, the error term can be decomposed as

$$\xi_{kt} = \xi_k + \xi_t + \tilde{\xi}_{kt},$$

where ξ_k and ξ_t denote model and market fixed effects, respectively. Identification relies on the assumption that observed characteristics are uncorrelated with the part of the error term that is not controlled for with fixed effects, $\tilde{\xi}_{kt}$.

A well-known issue in estimating demand systems is the endogeneity of prices and market shares, as prices are likely correlated with unobserved product characteristics and demand shocks. To address this, we employ instrumental variables that are correlated with prices but exogenous to $\tilde{\xi}_{kt}$. First, we use the lagged price of aluminum multiplied by a model's weight as a cost-shifter instrument, reflecting the intuition that heavier cars require more material and thus respond more to input cost changes. Second, we instrument the within-nest market share $\log(s_{kt|b})$ using the number of products within the same nest, as increased competition within a nest should mechanically reduce a given model's share.

Models are assigned to nests based on similarities in horsepower and weight. Using a k-means clustering algorithm, we create four nests and ensure that a model remains in the same nest across years.⁹ Summary statistics for the resulting nests are provided in Appendix C.

Estimation results

We show results from estimation of equation (9) in Table 2. Column (1) displays results from an OLS regression without instruments. Column (2) shows results from an IV regression using lagged aluminum prices multiplied with a model's weight as well as the number of products in the same nest as instruments. We include a model's horsepower-to-weight ratio (HP/W) and an indicator whether a model is observed for the first or last time in our data (to account for entry and exit) as characteristics.

The OLS regression serves as a valuable benchmark to evaluate the effectiveness of instrumental variables in the IV regression. The price coefficient α takes a value of -0.033 in the OLS regression. In the IV regression, the price coefficient α is about three times that magnitude and takes a value of -0.087, suggesting a higher price elasticity than column (1). We expect this shift as unobserved demand shocks correlated prices are likely to bias the price coefficient in the OLS regression towards zero. The coefficient on the within-nest market share λ is significant at the 5 percent level in the IV regression. The value of 0.2 suggests that there is substantial substitutability between products within the same nest. We also observe that models with a higher horsepower-to-weight ratio (HP/W) are associated with a higher market share relative to the outside option, holding all other characteristics fixed. A model that appears for the first or last time in our data (In/Out Indicator) has a lower market share relative to the outside option, holding all other characteristics fixed.

Columns (3) and (4) in Table 2 show the two first-stage regressions. Looking at the first stage for the price variable in column (3), we observe that a higher value of the lagged aluminum price is associated with a higher market share relative to the outside option. Thus, the direction of the correlation of our instrument and the price variable

9. We also experimented with fewer and more nests. Moving from three to four nests produced distinct clusters in terms of horsepower, weight, and prices, whereas adding a fifth nest resulted in two nest categories that were largely identical in terms of horsepower, weight, and prices. We therefore adopt four nests in the baseline specification.

is as expected. We can make a similar observation for the first stage of the log-market share within a nest. We observe that the larger the number of competing models within a nest, the lower the market share of a model within that nest, holding all else equal. F-Statistics of both first stages take values above 80.

Table 2: Nested logit estimation results

	(1) OLS	(2) IV	(3) 1st-stage: Price	(4) 1st-stage: Share
Price ('000£)	-0.033** (0.013)	-0.087*** (0.005)		
Log(Share within nest) (λ)	0.845*** (0.018)	0.196** (0.095)		
HP/W	-5.549* (3.098)	18.818*** (6.280)	322.897*** (69.009)	-2.433 (1.507)
In/Out Indicator	-0.374*** (0.038)	-1.779*** (0.201)	-0.175 (0.411)	-2.171*** (0.063)
Lagged aluminum price $\times$ Weight			0.013*** (0.001)	-0.001*** (0.000)
# Products in nest			-0.107*** (0.026)	-0.027*** (0.002)
Constant	-3.282*** (0.371)	-7.001*** (1.128)	-23.947** (12.075)	-0.924** (0.447)
R^2	0.840	0.441		
Year FE	Yes	Yes		
Brand FE	Yes	Yes		
First Stage F			418	83
Observations	5590	5590	5590	5590

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: all columns include make fixed effects. Standard errors are clustered at the model level. Prices are deflated using the CPI and expressed in Thousands of 2018 GBP. 'HP/W' refers to a model's horsepower-to-weight ratio. 'In/Out Indicator' is a dummy variable that equals one if a model is observed the first or last time in our data and zero otherwise in order to account for entry and exit.

Supply-side and markup formula

Estimates of the parameters α and λ , combined with an assumption on market conduct, allow us to identify markups. We assume that firms set prices under static, full-information Bertrand-Nash competition—a standard assumption in the demand estimation literature, particularly in the context of automobiles (e.g., Berry et al. 1995; Grieco et al. 2024). Under this assumption, the first-order conditions of profit maximization imply that the markup of model k in market t (suppressing the time subscript for notational convenience) can be expressed as

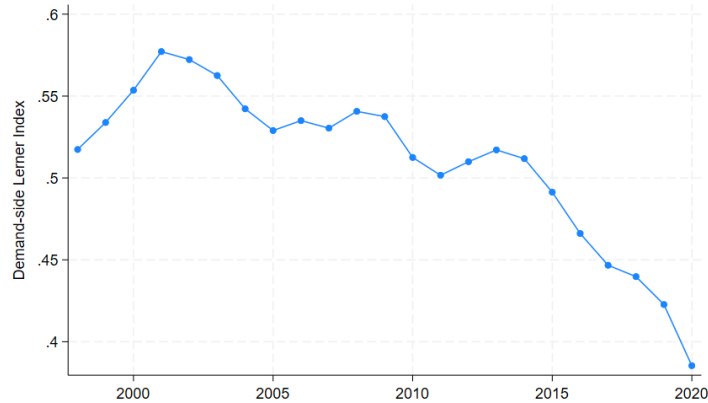
$$\frac{p_k - c_k}{p_k} = \frac{[-\Omega(p)^{-1}s(p)]_k}{p_k}, \quad (11)$$

where c_k denotes the marginal cost of producing model k , and $s(p)$ represents observed market shares. The matrix $\Omega(p)$ contains own- and cross-price derivatives with respect to models owned by the same firm in the same market.¹⁰ These derivatives are computed using the results from our nested logit estimation and equation (10). A full derivation of the markup formula is provided in Appendix E.

Figure 4 presents the revenue-weighted average demand-side Lerner index over time. Two patterns are particularly notable. First, markups are high in absolute levels, peaking at approximately 0.57 in 2001. Second, average markups decline steadily over the sample period, falling from 0.51 in 1998 to 0.39 in 2020—a reduction of almost 30%.

Figure 5 reports the demand-side Lerner index for selected manufacturers. Over time, markups generally decline for most makes. Cross-sectional differences are also pronounced when compared with production-based markups (Figure 3). For instance, Land Rover exhibits relatively low demand-side markups, ranging from 0.2 to 0.3, whereas Vauxhall’s demand-side markups exceed 0.6. Other manufacturers display similarly heterogeneous patterns.¹¹

Figure 4: Aggregate Demand Lerner index

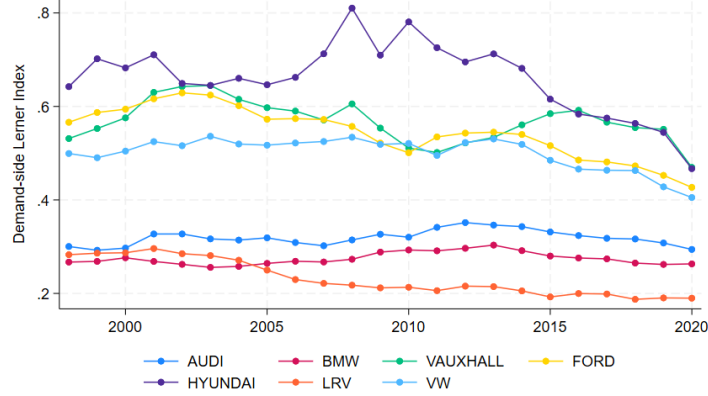


Note: figure shows the average revenue-weighted demand Lerner index.

10. Ownership is defined at the global level. For example, we treat the Volkswagen Group as the profit-maximizing entity for all its makes, including Audi, Skoda, and Volkswagen.

11. Demand and production markup estimates cannot be directly compared for all makes, as some vehicles sold in the UK are produced abroad and therefore do not appear in our production dataset.

Figure 5: Aggregate Demand Lerner index by Make



Note: figure shows demand-side Lerner index for selected makes, calculated as the revenue-weighted average over models.

3.2.3 Demand markups in the Rest of the World (ROW)

Viewed together, the results of the preceding subsections may appear contradictory: while production markups have doubled between 1998 and 2018, UK demand markups have fallen by almost one-third. One possible explanation is that the assumptions underlying one or both approaches are violated, rendering the respective markup estimates inconsistent. Alternatively, as suggested by the aggregation framework in Section 2, the divergence may be due to changes in the ROW demand markup.

To our knowledge, no prior study has attempted to quantify a ROW (i.e. “export”) demand markup.¹² If one had access to price and purchase data from all of the UK’s major car export destinations, it would be possible to use our proposed aggregation formula to test the assumptions underpinning the production and demand markup estimators. Such data are, however, generally unavailable. Instead, we assume that both production and demand markup estimation approaches are valid—that is, firms cost-minimize and set prices according to static Bertrand-Nash competition—and use this assumption to infer ROW demand markups. This allows us to assess how export markups can reconcile the divergent trends observed in production and UK demand markup estimates.

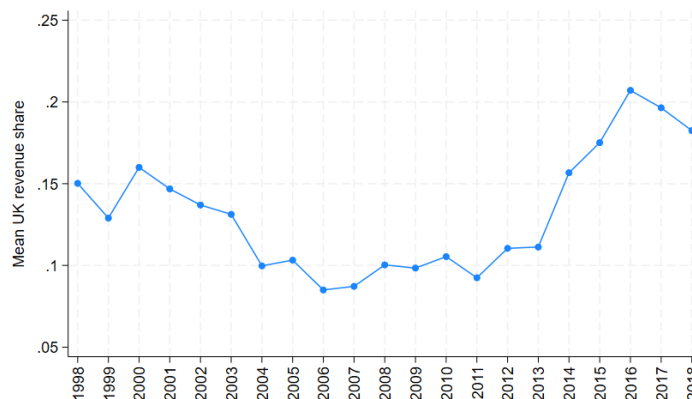
Inferring ROW demand markups requires firm-level information on export revenue. While we do not observe this directly, we approximate it as the residual between total annual revenues of UK car manufacturers (from the HO data) and the product of price and quantity for models manufactured and sold domestically (from the JATO data) in the same year. Although this approach may be subject to measurement error, for example due to inventory effects, we leverage additional export data from an industry organization

12. This is somewhat surprising given evidence that production markups tend to rise when firms begin exporting (e.g., De Loecker and Warzynski 2012).

and show in Appendix Figure F.1 that our estimated export revenue shares closely track observed export quantities.

Figure 6 plots the mean share of revenue generated from UK sales over time. Throughout the period, domestic sales account for only a small fraction of total revenues, implying that markups are predominantly determined by export demand conditions. Exports consistently represented at least four-fifths of total revenue over the two decades. The UK share fell from 15% in 1998 to under 10% in the early 2000s, before rising toward 20% in the end of the sample period.

Figure 6: UK revenue share



Note: figure shows mean UK revenue shares calculated by inferring UK revenue from JATO data and taking total revenue from HO.

Using the revenue share estimates, we invert equation (6) to recover the ROW Lerner index:

$$\hat{\mu}_{iROW}^L = \frac{R_i}{\hat{R}_{iROW}} \left(\hat{\mu}_i^L - \left(\frac{\hat{R}_{iUK}}{R_i} \right) \hat{\mu}_{iUK}^L \right). \quad (12)$$

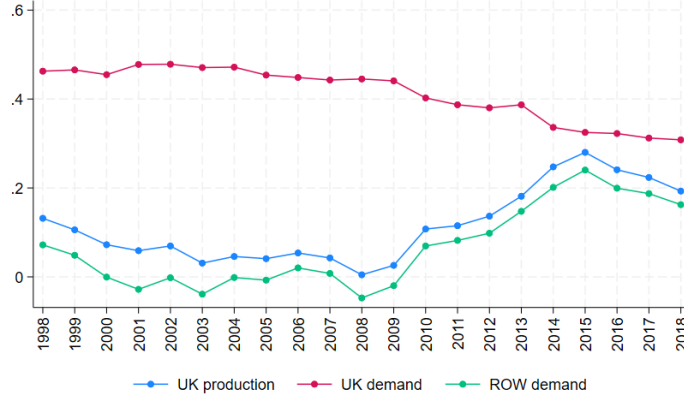
This expression allows us to compute the ROW markup necessary to reconcile production and UK demand markup estimates.¹³

Figure 7 plots markup estimates using the production and demand methods described in the preceding subsections as well as the ROW demand method outlined above. The contrast between the UK production and UK demand markups is stark when viewed together: the UK demand markup is more than three times as large at the beginning of the period and declines while the UK production markup increases. The ROW demand series shows what one must believe has happened to markups obtained by UK-based firms on their exports for both markup series to be consistent. Between 1998 and 2009 ROW

13. In subsection 3.3 use supplementary firm-level export data to assess the plausibility of the implied trends in ROW markups.

demand markups were essentially zero and even turned negative in the early 2000s.¹⁴ ROW markups have risen sharply since 2009, largely mirroring the upward trend in UK production markup, albeit from a lower base. The strong co-movement between UK production and ROW demand markups reflects the low share of UK revenues (Figure 6), implying that the production-based estimate is dominated by ROW conditions.

Figure 7: Aggregate Lerner index Markups



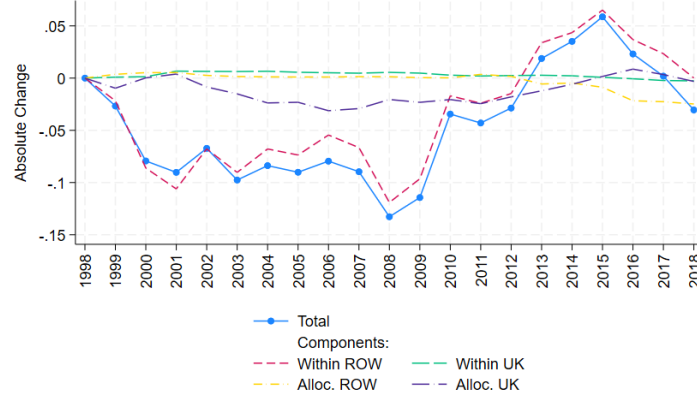
Note: figure shows revenue-weighted mean markup estimates. Series are restricted to car brands with manufacturing operations in the UK.

To what extent have changes in demand markups within markets and changes in market shares driven the UK production markup? To address this question, Figure 8 implements the decomposition in equation (5), separating the contributions of the UK and ROW markets. The figure plots the cumulative change in the UK production markup since 1998 alongside the cumulative contributions from the within- and between-market components. The decomposition reveals that the increase in the UK production markup over this period is entirely driven by within-market increases in the ROW markup. This upward effect is partially offset around the middle of the sample by a decline in the share of output sold in the UK, where demand markups are higher, and by a reduction in the UK demand-side markup.¹⁵

14. The negative Lerner index markup implies firms were losing money on exports, which could be rationalised by firms running losses in order to enter new markets.

15. The decomposition focuses on changes rather than levels. Since production-side markups are estimated using revenue data, one might worry about potential level biases (Bond et al. 2021). However, De Ridder et al. (2024) show that trends in production-side markups are identified using revenue data. Therefore, our main conclusion concerns differences in trends rather than levels, as illustrated in Figure 8.

Figure 8: Change in Aggregate Production Lerner index



Note: figure shows the cumulative absolute difference in the revenue-weighted mean producer-side Lerner markup estimate and its components.

In summary, if our estimates of both UK production and UK demand estimates are unbiased, their divergent trends can be reconciled by a marked increase in the ROW demand markup. In the following section we leverage administrative trade data to gauge whether such an increase is plausible.

3.3 Export patterns and markups

Our examination of the ROW demand markup thus far has been triangulation: given the UK demand markup, the UK production markup and the share of revenue UK manufacturers' gain from domestic sales, ROW demand markups can be inferred under the assumption that our estimates of UK production and demand markups are unbiased. If this is the case, Figure 8 shows car manufacturers' markups on their exports rose sharply between 2009 and 2018. If export destinations had remained fixed, such an increase would require implausibly large shifts in domestic demand. We therefore supplement our markup analysis with trade data to provide suggestive evidence on the plausibility of rising ROW markups.¹⁶

We combine administrative customs data from the Trade in Goods (TIG) data (Office for National Statistics 2022) with firm-level financial data from the Annual Business Survey (ABS) (Office for National Statistics 2023, 2024).¹⁷ This allows us to relate firm-level production markups to country-specific export shares.¹⁸ Focusing on major export destinations, we estimate the following specification for UK car manufacturers:

16. In principle, one could decompose the ROW demand markup into country-specific components by matching our UK demand estimates to administrative firm-level trade data. We are unable to do so since the data used to estimate UK demand markups lacks the anonymous firm identifiers required for administrative data matching.

17. We describe this data in detail in Appendix C.

18. Further details on both datasets and descriptive statistics are provided in Appendix C.

$$\hat{\mu}_{it} = \beta_0 + \beta_1 r_{itCN} + \beta_2 r_{itUS} + \beta_3 r_{itEU} + \beta_4 \mathbb{I}[X_{it}] + \tau_t + \epsilon_{it} \quad (13)$$

where $r_{itCN/US/EU}$ are Chinese/US/EU export revenue shares defined as firms' value of exports to the specific destination divided by revenue. $\mathbb{I}[X_{it}]$ is an indicator function that takes the value of 1 if a firm exports and zero otherwise.

Table 3: Markups and Export Markets

	(1)	(2)	(3)	(4)
China turnover share	0.447*** (0.127)	0.427*** (0.143)	0.457* (0.245)	0.329** (0.146)
US turnover share	0.318*** (0.104)	0.313*** (0.110)	0.094 (0.147)	0.055 (0.136)
EU turnover share	-0.075* (0.042)	-0.083 (0.061)	-0.076 (0.131)	-0.130 (0.099)
Exporter dummy	-0.113*** (0.036)	-0.150 (0.119)	-0.061 (0.040)	-0.984*** (0.358)
Ln(Exports)		0.002 (0.006)		0.053*** (0.020)
Year dummies	Yes	Yes	Yes	Yes
Firm FEs	No	No	Yes	Yes
N obs.	387	387	387	387
N firms	107	107	107	107

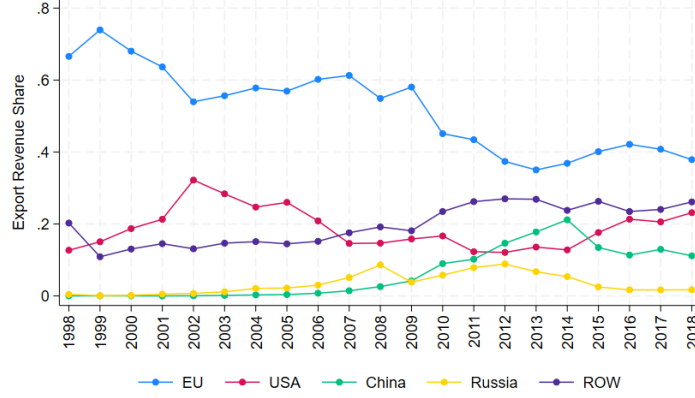
Note: standard errors in parentheses clustered at the firm level. */**/** denote significance at the 10/5/1 percent level respectively. Estimates weighted by revenue. Dependent variable is firm-level markup in year t .

Column (1) of Table 3 reports estimates of equation (13), while column (2) additionally controls for the log of total exports. Columns (3) and (4) replicate these specifications with the inclusion of firm fixed effects. Across all specifications, exporting to China and the US is positively associated with markups. The effect of exports to China is consistently large and statistically significant, whereas the association with US exports is smaller and becomes insignificant once firm fixed effects are included. In contrast, exporting to the EU exhibits a zero or negative relationship with markups, potentially reflecting stricter EU competition policy during this period, which aimed to reduce price differentials across Member States.¹⁹

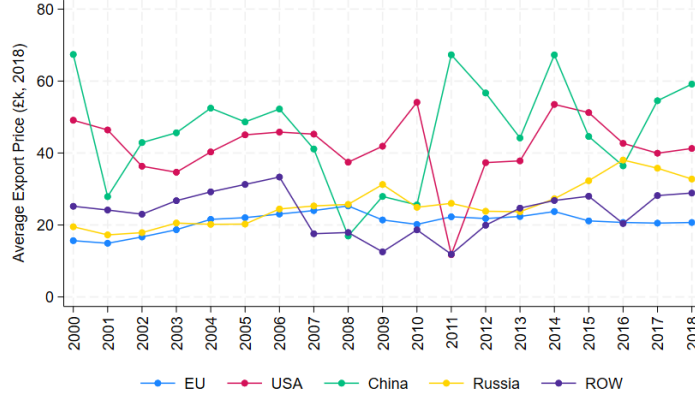
19. See, for example, the EU Commission's [competition policy documents on motor vehicles](#).

Figure 9: Export Characteristics

(a) Export Markets



(b) Average Prices



Note: figure 9a shows shares of total UK car export revenue by destination. Figure 9b shows average prices of UK car exports by destination, calculated as the value of exports divided by the number of units exported. Cars identified as HS code 8703 ‘Motor cars and other motor vehicles principally designed for the transport of <10 persons’. Data from UN Comtrade.

The results in Table 3 indicate that firms’ markups vary across export destinations, suggesting substantial differences in consumer preferences across markets. The strong positive association between firm-level markups and exports to China and the US implies that expansion into these markets may account for the large increase in ROW demand markups documented in Section 3.2.3. Figure 9a shows that China and the US are precisely the markets that have expanded since 2009, coinciding with the period of rising ROW markups. Figure 9b further shows that average export prices (unit values) to China and the US are substantially higher than those to the EU and have increased, albeit with some volatility, since 2009.

Overall, these findings corroborate the increase in export markups documented in

Section 3.2.3 and suggest that the sharp rise in ROW markups for UK car manufacturers since 2009 is both plausible and primarily driven by expansion into the relatively high-markup markets of China and the US.

4 Conclusion

Demand- and production-based approaches to markup estimation rely on different data and assumptions. Demand-based methods model consumer choice and firm pricing behavior, while production-based approaches assume cost minimization and flexible input use. We show that even when both sets of assumptions hold, demand and production markup estimates can diverge when firms sell in multiple markets. To formalize this, we derive an aggregation formula demonstrating that production markups can be expressed as a revenue-weighted average of market-specific markups. This approach offers a practical advantage over cost-share-based aggregations (De Loecker and Warzynski 2012; Cairncross et al. 2023), as revenue shares are typically observable, whereas cost shares often require additional assumptions about production technology.

We illustrate the relevance of this framework in the context of the car market - an industrial context that has long been a laboratory for markup estimation. We find demand markups *declined* between 1998 and 2018, while production markups *increased* over the same period due to an increase in the average markup that UK car manufacturers gain on their exports. We corroborate this finding using administrative trade data, which suggests expansion into China where luxury brand markups are high has driven the increase in production markups.

Our theoretical and empirical results show demand and production markups are conceptually different quantities in the presence of trade. This work highlights that future research leveraging price and purchase data across major export markets would be particularly valuable, as it would allow the estimation of country-specific elasticities and markups thus enabling a test of the assumptions underlying production and demand estimation approaches. Combining such demand-side data with firm-level cost and trade information would shed light on how firms allocate pricing power across domestic and foreign markets, how international competition shapes markups, and how consumer heterogeneity influences pricing strategies. These insights would clarify the relationship between production and demand-based markup estimates in open economies and offer a richer framework for understanding firm behavior and market power in global markets.

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Online Appendices

A Price-marginal cost ratio aggregation

Consider price-marginal cost (PMC) markups

$$\mu_{ijk}^{PMC} = \frac{p_{ijk}}{c_{ijk}}, \quad (14)$$

where all notation is as before. As noted by De Loecker and Warzynski (2012) and Cairncross et al. (2023), one can express aggregate firm-level PMC markups as a *cost share*-weighted average of market or product-specific PMC markups (De Loecker and Warzynski (2012) focus on aggregation across markets, while Cairncross et al. (2023) focus on aggregation across products). However, this requires data on the distribution of aggregate firm costs across goods or markets, which is not typically available. Alternatively, one can also express aggregate PMC markups as a revenue-weighted function of model-market PMC markups.

For conciseness, we drop the subscript k and ignore aggregation across goods.²⁰ As De Loecker (2011) highlights, firm i 's PMC markup can be expressed as

$$\mu_i^{PMC} = \theta_i^X \frac{R_i}{C_i}, \quad (15)$$

where θ_i^X denotes the output elasticity of a perfectly variable input X for firm i .²¹ This can be interpreted as a revenue-weighted function of μ_{ij}^{PMC} since

$$\begin{aligned} \mu_i^{PMC} &= \theta_i^X \frac{R_i}{C_i} \\ &= \theta_i^X R_i \left(\sum_j C_{ij} \right)^{-1} = \left(\frac{1}{\theta_i^X R_i} \sum_j C_{ij} \right)^{-1} \\ &= \left(\sum_j \frac{R_{ij}}{R_i} \frac{C_{ij}}{\theta_i^X R_{ij}} \right)^{-1} = \left(\sum_j \frac{R_{ij}}{R_i} \frac{1}{\mu_{ij}^{PMC}} \right)^{-1}, \end{aligned} \quad (16)$$

where the final equality makes the assumption that the output elasticity of a static input

20. The approach to aggregating across goods is analogous to aggregating across markets, as demonstrated in sub-section 2.

21. More precisely, ‘perfectly variable’ in this context means an input subject to zero adjustment costs and which can therefore be optimised period-by-period. Conditional on other dynamic inputs, such as capital, this allows one to ignore dynamic aspects of a firms’ cost-minimisation problem, which leads to the neat closed-form markup expression of equation 15.

X is constant across markets.²² Inverting the Left Hand Side of equation (16) gives

$$\frac{1}{\mu_i^{PMC}} = \sum_j \frac{R_{ij}}{R_i} \frac{1}{\mu_{ij}^{PMC}}, \quad (17)$$

which shows the inverse of a firm's PMC markup can be expressed as a revenue-weighted average of the inverse of their market-specific PMC markups. Although less straightforward than the Lerner-index aggregation, it is reassuring that one can express both Lerner-index and PMC markups as revenue-weighted averages of market-specific markups.

B Non-constant marginal costs

The derivations in the main text assume marginal costs are constant with respect to the quantity produced, i.e. $MC(q) = \bar{c}$. This assumption implies the Cost of Goods Sold (COGS) can be written as the product of quantity and marginal cost ($COGS = q \cdot \bar{c}$). Accordingly, with constant marginal costs, markups can be directly computed from the observation of revenue and COGS.²³ The markup estimation approach of De Loecker and Warzynski (2012) does not require this assumption. Instead, marginal costs are treated as unobserved and markups are backed out structurally from the estimated production function in combination with assuming cost-minimising firm behaviour. We show that we can derive a similar markup aggregation formula as in the main text in the case of non-constant marginal costs.

For simplicity, focus on the case where the firm sells a product either into domestic or foreign markets. The quantity sold is q^D for the domestic market and q^F for the foreign market. Similarly, the price differs between markets and is denoted as p^D and p^F . Firms produce the total quantity $q = q^D + q^F$ and the marginal cost of producing an additional unit is $c = MC(q)$. The average price which the firm obtains on its sales is $p = \omega^D p^D + \omega^F p^F$, where $\omega^D = \frac{q^D}{q}$ is the fraction of the quantity which is sold in the domestic market (similarly for ω^F). The revenue which the firm receives in market k is $R^k = p^k q^k$ and total revenue is $R = R^D + R^F$.

The aggregate markup from the perspective of the firm compares the average price it achieves on its sales with the cost of producing an additional unit:

$$\mu = \frac{p - c}{p} = \frac{\omega^D p^D + \omega^F p^F - c}{\omega^D p^D + \omega^F p^F}$$

22. This assumption may be violated if, for example, there were substantial variation in quality across markets or if firms manufactured multiple products, each with different output elasticities, and sold products in different ratios across markets.

23. Since revenue is the product of quantity and price ($R = q \cdot p$), the ratio of COGS and revenue gives the price-cost margin with constant marginal costs ($\frac{R}{COGS} = \frac{p}{\bar{c}}$).

This expression can be rearranged:

$$\begin{aligned}
\mu &= \frac{\omega^D p^D + \omega^F p^F - c}{\omega^D p^D + \omega^F p^F} \\
&= \frac{\omega^D p^D + \omega^F p^F - \omega^D c - \omega^F c}{\omega^D p^D + \omega^F p^F} \quad (\text{using } c = \omega^D c + \omega^F c) \\
&= \frac{\omega^D p^D - \omega^D c}{\omega^D p^D + \omega^F p^F} + \frac{\omega^F p^F - \omega^F c}{\omega^D p^D + \omega^F p^F} \\
&= \frac{q^D p^D - q^D c}{q^D p^D + q^F p^F} + \frac{q^F p^F - q^F c}{q^D p^D + q^F p^F} \quad (\text{multiplying by } \frac{q}{q}) \\
&= \frac{q^D p^D - q^D c}{q^D p^D + q^F p^F} + \frac{q^F p^F - q^F c}{q^D p^D + q^F p^F} \\
&= \frac{R^D}{R^D} \frac{q^D p^D - q^D c}{R} + \frac{R^F}{R^F} \frac{q^F p^F - q^F c}{R} \\
&= \frac{R^D}{R} \frac{q^D p^D - q^D c}{R^D} + \frac{R^F}{R} \frac{q^F p^F - q^F c}{R^F} \\
&= \frac{R^D}{R} \frac{p^D - c}{p^D} + \frac{R^F}{R} \frac{p^F - c}{p^F} \\
&= \frac{R^D}{R} \mu^D + \frac{R^F}{R} \mu^F
\end{aligned}$$

μ^D is the domestic markup, which compares the domestic price of a unit (p^D) to the marginal cost of producing an additional unit ($c = MC(q)$) and similarly μ^F is the foreign markup. As in the main text, the aggregate markup is the revenue-weighted sum of these market-specific markups.

C Data

C.1 Historical Orbis

Historical Orbis (HO) includes data on active and dead incorporated firms in various corporate databases constructed by Bureau Van Dijk (BVD). These databases compile accounting information on firms as well as attributes such as location and industry from corporate documents such as annual reports and account filings. The UK component of this is FAME and spans several decades. An earlier iteration of FAME, known as Amadeus, included observations on some notable UK car manufacturers which are absent in the latest data. We therefore combine HO with Amadeus in an effort to improve the coverage of our corporate financial data.

HO often contains multiple observations per firm-year pertaining to different information sources. We obtain a single observation per firm-year by prioritising information from annual reports over local registry filings, information filed at the end of a month rather than the beginning and consolidated over unconsolidated accounts. We harmonise data to account for differences in reporting period length and span by assuming values

are distributed uniformly across months in a given period, this allows us to expand the reporting periods into a monthly panel, which we use to construct observations at the calendar year-firm level.

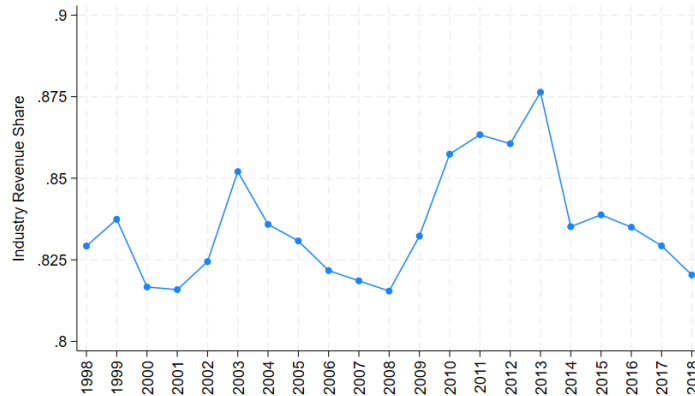
HO contains multiple industry codes per firm to reflect a range of their activities. We classify firms as car manufacturers if *any* of their associated NAICS codes are recorded as 336111 (‘Automobile Manufacturing’). Our analysis sample is restricted to car manufacturers that are observed in the JATO data. Table C.1 shows the characteristics of this sample compared to all car manufacturers observed in HO. The average firm in our analysis sample is clearly far larger than the average car manufacturer observed in the UK, which is due to the JATO data being restricted to car manufacturers with non-negligible shares of the UK market. Figure C.1 shows that these firms account for the vast majority of revenue in the industry.

Table C.1: Historical Orbis Descriptive Statistics

	UK Car Manufacturers					
	All			Analysis Sample		
	Mean	Median	S.D.	Mean	Median	S.D.
Revenue (£bn)	0.43	0.01	2.01	6.81	4.59	5.71
Employment (ths)	0.6	0.1	2.4	7.4	4.6	7.2
Wage bill (£bn)	0.06	0.01	0.24	0.60	0.35	0.64
Material Inputs (£bn)	0.58	0.02	1.91	5.42	3.97	3.63
Fixed assets (£bn)	0.04	0.00	0.50	1.96	0.99	3.19
EBITDA (£bn)	0.01	0.00	0.20	0.21	0.12	0.80
N Obs.	13873			167		
N Firms	1962			10		

Note: monetary values are deflated using the CPI and expressed in 2018 prices.

Figure C.1: Revenue Share of Sample Manufacturers



Note: figure shows the share of total revenue among UK car manufacturers accounted for by manufacturers in our analysis sample.

C.2 Demand-side data

Our primary source of demand-side data was obtained from JATO, a market intelligence firm. This data includes price, weight, horsepower, sales and number of trims at the year-make-model-country of assembly level. The data provided is averaged across number of trims (a model can be sold in different versions, referred to as *trim*), weighted according to trim-specific sales.

The measure of price observed in the JATO data is the price suggested by the manufacturer including VAT. We thus do not observe actual transaction prices, i.e. those that include potential discounts granted by dealers. For newly introduced models, we do not observe the month of first listing. Similarly, for models that stopped being sold, we do not observe the exact month a model is taken out of the market. This may cause a problem if entering/exiting firms have a low market share because they were not sold for an entire year. We account for this potential problem by including a dummy variable indicating the first/last time a model is observed.

We complement the JATO registration data with various other data. These include: Consumer Price Index (CPI) data from ONS, data with the aggregate number of households in the UK by year from ONS, data on real exchange rate relative to the US in different countries from Penn World Tables (PWT), average annual aluminium price data from St. Louis FED (FRED), and UK income data from the Family Resources Survey (FRS).

Figure C.2: Number of Registrations



Note: Number of registrations by year published by UK Driver and Vehicle Licensing Agency (DVLA) compared with JATO data. DVLA data only available 2001 onwards.

Figure C.2 compares the number of registrations observed in the JATO data with official data published by the UK Driver and Vehicle Licensing Agency (DVLA). We observe a slight discrepancy at the beginning of the time period, but numbers and trends are very similar. The slight discrepancy arises because JATO provides us with data of models that are within the top 90% of sales in at least one year of the time period (i.e. if a model is within the top 90% in 2005 but in no other year, it will show up in the data for all years of the time period 1998-2022).

Table C.2 shows summary statistics of our sample. The number of brands has stayed relatively constant, registrations fluctuate with the business cycle, and the shares of cars produced in Great Britain and the European Union have decreased, respectively. Average real prices have seen an increase over time as well as the observed average horsepower to weight ratio.

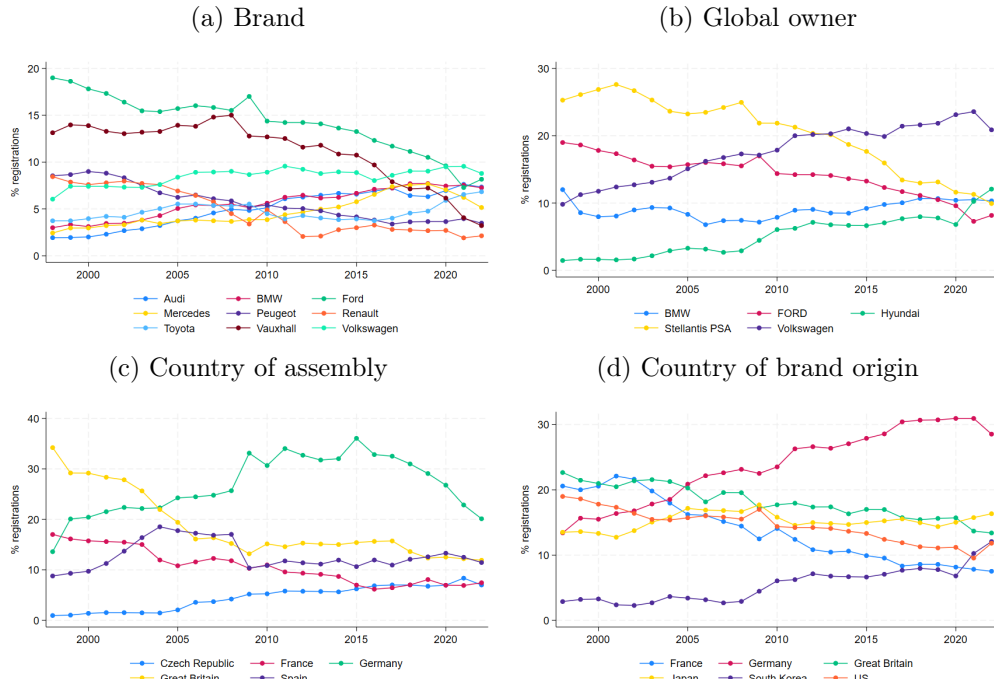
Table C.2: Descriptive Statistics of Demand Data

Year	Models	Brands	Registrations	GB produced	EU produced	GB brand	Price	HP/W	Trims
1998	173	25	2.123	0.34	0.89	0.23	21.159	0.084	9.96
1999	179	25	2.083	0.29	0.88	0.21	20.868	0.085	9.38
2000	193	25	2.100	0.29	0.90	0.21	20.152	0.085	9.05
2001	197	26	2.332	0.28	0.91	0.20	19.376	0.086	9.25
2002	200	26	2.443	0.28	0.91	0.21	19.623	0.087	8.64
2003	204	26	2.449	0.26	0.89	0.22	20.129	0.086	8.44
2004	210	26	2.385	0.22	0.86	0.21	20.878	0.086	8.58
2005	213	26	2.211	0.19	0.86	0.20	21.482	0.086	9.00
2006	220	25	2.128	0.16	0.87	0.18	21.591	0.087	8.49
2007	227	25	2.204	0.16	0.90	0.20	21.735	0.089	8.45
2008	233	24	1.988	0.15	0.90	0.20	21.154	0.089	8.69
2009	240	24	1.859	0.13	0.88	0.17	21.057	0.088	8.86
2010	240	25	1.946	0.15	0.88	0.18	22.132	0.089	8.61
2011	234	25	1.866	0.15	0.89	0.18	22.623	0.091	9.45
2012	248	25	1.973	0.15	0.88	0.17	22.342	0.091	9.30
2013	250	25	2.199	0.15	0.88	0.17	22.161	0.092	9.45
2014	250	26	2.386	0.15	0.88	0.16	22.485	0.093	9.14
2015	255	26	2.507	0.15	0.88	0.17	23.541	0.096	9.90
2016	254	26	2.571	0.16	0.86	0.17	24.829	0.099	9.69
2017	251	26	2.422	0.16	0.85	0.16	25.827	0.101	9.90
2018	254	26	2.236	0.14	0.84	0.15	26.211	0.103	9.29
2019	262	26	2.200	0.12	0.84	0.16	27.053	0.104	9.48
2020	243	26	1.550	0.13	0.83	0.16	29.047	0.109	9.21
2021	245	26	1.540	0.12	0.79	0.14	29.895	0.111	9.02
2022	251	26	1.495	0.12	0.75	0.13	29.604	0.114	7.91

Note: columns 2 to 4 show count by year in each cell. Columns 5 to 7 show the share of registrations produced in GB/EU and the share of registrations from GB brands. Columns 8 to 10 show registration weighted means in each cell. Registrations are shown in Millions and Price in Thousand pounds deflated to 2018 price level. HP/W refers to the horsepower to weight ratio. A car is classified as EU produced if the country of assembly was a member of the EU in at least one year over the entire sample period (e.g. Croatia only joined the EU in 2013, but cars produced in Croatia prior to 2013 are classified as EU produced). GB brand refers to the share of cars produced by firms with British origin, e.g. Vauxhall (although Vauxhall is owned by a foreign company).

Figure C.3 shows additional trends in the data over time. The four panels show the market shares by brand, global owner, country of assembly, and country of brand origin for our sample period.

Figure C.3: Market Shares



Note: figures show selected brands/countries (lines do not add up to %). Country of assembly is a variable provided by JATO. Country of brand origin is self-coded. For example, Vauxhall is coded as British, Volkswagen as German, and Skoda as Czech (note that it is about the origin of the brand, not about ownership or production).

UK revenue shares are inferred from JATO data by selecting models that are recorded as UK-made and calculating UK revenue = MSRP*Volume. The following makes in the JATO data are combined as they appear under one manufacturer in the UK data.

- Renault is combined with Nissan as they operate as a single entity known as the 'Renault-Mitsubishi Alliance'.
- Mazda is combined with Ford as the latter owned the former in the years that Mazda is observed as UK-made (1998-2002).
- Land Rover is combined with Jaguar from 2013 onwards as Land Rover's revenue as recorded in HO is negligible after this date.
- MG is combined with Rover during its years of operation, as the two manufacturers were co-owned during their years of operation (1998-2004).

C.3 Annual Business Survey

The Annual Business Survey (ABS) is an annual survey conducted by the Office for National Statistics and regarded as the principal source of UK business data. Before

2009, it was referred to as the Annual Business Inquiry – part 2 (ABI/2), with historical data stretching back to 1995 for the entire economy and to 1980 for manufacturing.

The ABS collects information on key financial metrics including revenue, purchases, employment costs and capital expenditure from companies in the UK’s non-financial business sector, which constitutes roughly two-thirds of the entire UK economy. Approximately 62,000 businesses across Great Britain and approximately 11,000 businesses in Northern Ireland are sampled from the UK’s Inter-Departmental Business Register (IDBR), which includes information on companies registered for Value Added Tax (VAT) and employee tax, as well as incorporated businesses registered at Companies House. The IDBR includes around 2.6 million businesses accounting for nearly 99% of UK economic activity although it excludes the self-employed without employees not registered for employment tax, companies with minimal revenue not registered for VAT, and some non-profit organizations. The ABS utilizes a stratified random sample design with sampling strata based on employment size, industry code and geographical region. Firms with at least 250 employees are sampled every year smaller firms sampled on a rolling basis such that a firm is generally selected for a two-year period and is then unlikely to be re-selected for at least another two years.

ABS questionnaires are sent to business entities known as ‘reporting units’. For most businesses, the reporting unit is typically the legal entity that has been registered for VAT and/or employment tax, or the incorporated entity registered at Companies House. In large, complex businesses or conglomerates that consist of multiple operations or divisions, the reporting unit might be a specific part of the business that is responsible for its own financial transactions and records. Reporting units are assigned a single industry code according to the activity that generates the majority of their revenue. We classify firms as car manufacturers if this code is recorded as Standard Industrial Classification code 2901 (‘Manufacture of motor vehicles’). Although the manufacture of parts and accessories for motor vehicles are recorded under a different industry code, it should be noted that SIC 2901 spans more than the manufacture of finished vehicles, including operations such as the manufacture of vehicle engines and chassis.

We combine the ABS with the Trade in Goods data described below to measure firms’ revenue and relate export revenue shares to production markups, estimated using the formula outlined in section 3.2.1. Summary statistics of our analysis sample are provided in the following subsection.

C.4 Trade in Goods

Trade in Goods (TIG) is administrative data drawn from trade declarations submitted to HMRC, the UK’s tax authority aggregated to the firm-country-product-flow-year level. Prior to the UK’s departure from the European Union in 2020, trade with EU member countries only had to be reported if the annual value of flows was above a certain threshold. These thresholds apply to imports and exports separately and have changed over time starting at £225k for both imports and exports at the start of the sample period in 2005 but diverging subsequently with the threshold for imports increasing by far more than the threshold for exports. In 2019, the threshold for reporting imports was £1.5mn

while for exports it was £250k. This means the coverage of exports to the EU will be more comprehensive than the coverage of imports from the EU in later years. Trade flows to and from non-EU countries are reported irrespective of value.

We match TIG and the ABS using unique reporting unit identifiers and use the TIG information to calculate export revenue shares. Summary statistics of our analysis sample are provided in table

Table C.3: Annual Business Survey-Trade in Goods Descriptive Statistics

	2005-2019		2005-2009		2010-2014		2015-2019	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Production Markup	1.10	0.14	1.10	0.14	1.09	0.15	1.09	0.13
Exporter Dummy	0.91	0.28	0.93	0.26	0.93	0.25	0.88	0.32
Ln(Exports)	15.73	5.88	15.60	5.49	16.52	5.42	15.22	6.44
China Export Turnover Share	0.02	0.05	0.01	0.02	0.03	0.07	0.02	0.04
US Export Turnover Share	0.05	0.12	0.04	0.08	0.04	0.10	0.06	0.15
EU Export Turnover Share	0.14	0.15	0.14	0.16	0.15	0.16	0.12	0.14
N obs.	430	430	129	129	130	130	171	171

Note: sample restricted to car manufacturers identified as SIC code 2901. Statistics weighted by revenue and ABS sampling weights.

D Production markup estimation

Table D.1: Dynamic Panel Production Function Estimates MA(0) Assumption

	(1)	(2)	(3)	(4)
	OLS (full samp)	OLS (MA(0) samp)	A-H MA(0)	B-B MA(0)
Ln(Materials)	0.905*** (0.051)	0.908*** (0.040)		0.846*** (0.058)
Ln(Wagebill)	0.060 (0.054)	0.064 (0.068)		0.102** (0.034)
Ln(Fixed assets)	0.076 (0.046)	0.075 (0.051)		-0.002 (0.031)
Delta Ln(Materials)			0.870*** (0.041)	
Delta Ln(Wagebill)			0.106*** (0.037)	
Delta Ln(Fixed assets)			-0.026 (0.029)	
Lagged Ln(Revenue)				0.559*** (0.096)
Lagged Ln(Materials)				-0.488*** (0.094)
Lagged Ln(Wagebill)				-0.043 (0.045)
Lagged Ln(Fixed assets)				0.038 (0.026)
N obs.	165	143	143	143
N firms	10	10	10	10
Adj. R2	0.974	0.970	0.921	
Wald p-val for CRS	0.283	0.224	0.291	
Hansen-J p-val			.	1.000
F statistic			7.12	29.21

Note: standard errors in parentheses clustered at the firm level. ‘Delta’ denotes 1-period differences and ‘Lagged’ denotes 1-period lags. */**/** denote significance at the 10/5/1 percent level respectively.

Table D.2: Dynamic Panel Production Function Estimates MA(1) Assumption

	(1) OLS (full samp)	(2) OLS (MA(1) samp)	(3) A-H MA(1)	(4) B-B MA(1)
Ln(Materials)	0.905*** (0.051)	0.899*** (0.036)		0.863*** (0.052)
Ln(Wagebill)	0.060 (0.054)	0.055 (0.067)		0.119** (0.045)
Ln(Fixed assets)	0.076 (0.046)	0.088 (0.051)		0.006 (0.026)
Delta Ln(Materials)			0.870*** (0.057)	
Delta Ln(Wagebill)			0.122*** (0.039)	
Delta Ln(Fixed assets)			0.022 (0.032)	
Lagged Ln(Revenue)				0.580*** (0.104)
Lagged Ln(Materials)				-0.517*** (0.102)
Lagged Ln(Wagebill)				-0.069 (0.045)
Lagged Ln(Fixed assets)				0.032 (0.027)
N obs.	165	132	132	132
N firms	10	10	10	10
Adj. R2	0.974	0.970	0.922	
Wald p-val for CRS	0.283	0.281	0.622	
Hansen-J p-val			.	1.000
F statistic			5.52	220.54

Note: same as note to table D.1.

Table D.3: Production Function Coefficients Implied by Blundell-Bond Results

	ARMA(1,0)	ARMA(1,1)
Implied coefficients		
Ln(Materials)	0.878***	0.866***
Ln(Wagebill)	0.079***	0.080**
Ln(Fixed assets)	-0.015	0.002
Lagged Ln(Revenue)	0.648***	0.756***
Comfac	0.032	0.156

Note: same as note to table D.1.

E Demand-side markup estimation

E.1 Derivation of demand-side markup formula

Assume that firms statically compete in prices under full information (Bertrand Nash competition) in each year. A firm f that produces a set of model F_f in market t with market share $s_k(p_k)$ of model $k \in K$ at price p_k , faces marginal cost c_k and pays fixed cost C_f , then maximises profits from model k Π_f with respect to price (dropping the market subscript t):

$$\max_{p_k} \Pi_f = \sum_{k \in F_f} (p_k - c_k) s_k(p_k) - C_f.$$

This yields the following set of FOCs for all $k \in K$:

$$\frac{\partial \Pi_f}{\partial p_k} = s_k(p_k) + \sum_{r \in F_f} (p_r - c_r) \frac{\partial s_r(p_r)}{\partial p_k} = 0.$$

Rearranging and stacking together FOCs across all models and firms gives (in vector/matrix notation):

$$\underset{K \times 1}{c} = \underset{K \times 1}{p} + \underset{K \times K}{\Omega(p)^{-1}} \underset{K \times 1}{s(p)}$$

where $\Omega(p)$ is a $K \times K$ matrix with cell-entries corresponding to derivatives with respect to the market share if product j and r are produced by the same firm, i.e.

$$\Omega_{rk}(p) = \begin{cases} \frac{\partial s_r(p)}{\partial p_k} & \text{if } r, k \in F_f \\ 0 & \text{otherwise.} \end{cases}$$

Inverting gives an expression for markups defined as the Lerner index:

$$\frac{p_k - c_k}{p_k} = \frac{[-\Omega(p)^{-1} s(p)]_k}{p_k},$$

where $[-\Omega(p)^{-1} s(p)]_k$ denotes the k^{th} element of the $K \times 1$ vector $[-\Omega(p)^{-1} s(p)]$.

Alternatively, the price cost ratio is:

$$\frac{p_k}{c_k} = \frac{p_k}{p_k + [-\Omega(p)^{-1} s(p)]_k}$$

Prices p_k and market shares $s_k(p)$ are observed in the data. Elements of the matrix $\Omega(p)$ are obtained from our estimates of demand parameters.

E.2 Additional demand estimation results

In addition to the nested logit model, we considered a random coefficient model following Berry et al. (1995). As discussed in the main part of the paper, this gives us insignificant random coefficient estimates. We therefore base demand-side markup estimates in the

main part of the paper on our nested logit model. We present the BLP-style model and the results in the following.

In this appendix, we consider the following demand system. In each year t households (i) decide whether to buy a car model k or choose the outside option ($k = 0$). Households maximise their conditional indirect utility U_{ikt} :

$$U_{ikt} = \beta_{it}x_{kt} + \alpha_{it}p_{kt} + \xi_{kt} + \epsilon_{ikt},$$

where x_{kt} denotes a vector of characteristics of model k in year t , p_{kt} is the price of model k in year t , ξ_{kt} is an unobserved demand shifter, and ϵ_{ikt} is demand shock following a Type-1 extreme value distribution (i.i.d. across products). Since we observe data aggregated at the UK level, we define markets as respective years t . We define the market share of product k in market t as number of sales divided by the number of households in the UK in year t . The outside option (i.e. not buying a new car) accounts for the largest market share in all years of our sample (around 0.9).

To allow for heterogeneity across consumers in their valuation of attributes we interact draws from the UK income distribution²⁴ and unobserved preferences with prices and car characteristics. Individual coefficients take the following form:

$$\begin{pmatrix} \alpha_{it} \\ \beta_{it} \end{pmatrix} = \begin{pmatrix} \bar{\alpha} \\ \bar{\beta} \end{pmatrix} + \Pi D_{it} + \Sigma v_{it}, \quad (18)$$

where D_{it} is income draw of individual i in year t and values v_{it} are drawn from a standard normal distribution. We draw income from the UK Family Resources Survey (FRS) using sampling weights, and demean as well as divide values by the observed standard distribution. In practice we restrict some of the elements in Π and Σ to zero, i.e. we do not interact every characteristics with income/unobserved heterogeneity. That is, in our main specification we interact prices with income, and interact draws from the normal distribution with the constant and horsepower-to-weight ratio. The inclusion of consumer heterogeneity through draws from observed and unobserved distributions should allow for more realistic substitution patterns. Nevo (2000) outlines market share formulas and expressions for elasticities in detail. In the interest of brevity we do not show these formulas here and refer the reader to the summary in Nevo (2000).

With the vector of instruments Z_{kt} at hand we can construct the following set of moments across all models k in each market t to estimate our vector of parameters $\theta = (\alpha, \beta, \Pi, \Sigma)$:

$$\mathbb{E} [\xi_{kt}(s_{kt}, p_{kt}, x_{kt}; \theta) Z_{kt}] = 0.$$

The key identifying assumption is that the unobserved demand shock ξ_{kt} is uncorrelated with our vector of instruments Z_{kt} .

We estimate the model using the PyBLP package.²⁵ The main estimation technique builds on Berry et al. 1995, and the PyBLP package allows us to implement many of

24. We use data from the Family Resources Survey (FRS) to generate the income draws.

25. See Conlon and Gortmaker (2020, 2023) for more details. A detailed documentation of PyBLP can be found here: <https://pyblp.readthedocs.io/en/stable/>.

the best practices pointed out by more recent literature. We simulate markets consisting of 1,000 individuals, using Modified Latin Hypercube Sampling (MLHS) following Hess et al. (2006) to draw from the standard normal distribution.²⁶ We impose a tight convergence criterion ($1e^{-12}$), and use the L-BFGS algorithm.

Table E.1: Random coefficient model estimates

Variable	$\bar{\alpha}, \bar{\beta}$ (Means)	Σ (Std deviations)	π (Income interactions)
Prices	-0.108 (0.05)	0.001 (0.21)	-0.002 (0.03)
Hp/Weight	28.958 (15.29)	— —	— —
Entry/Exit	-2.204 (0.08)	— —	— —
Time Trend	-0.052 (0.03)	— —	— —
Recession	-0.609 (0.19)	— —	— —
Constant	— —	— —	1.905 (0.95)

Note: Results from our baseline random coefficients model. The first column reports estimates of linear coefficients (means). The second column shows estimated coefficients from interactions with draws from a standard normal distribution (which can be interpreted as standard deviations from the mean). The third column shows estimated coefficients from interactions with income draws. The specification includes make fixed effects and standard errors are clustered at the model level. Number of observations is 5590.

Table E.1 shows results of the full random coefficients model. The vector of characteristics from equation x_{kt} contains the horsepower-to-weight ratio, an indicator whether a model k is observed for the first or last time in our sample in year t , a time trend, and a dummy variable for recessions. We additionally include make fixed effects to account for unobserved characteristics that are constant within a make across years (e.g. the reputation of a make). We instrument for prices using the lagged real exchange rate relative to the UK following Grieco et al. (2024).

The linear term on prices is negative and significant with a value of -0.11. We include an interaction of prices with draws from the standard normal distribution and interactions of prices and the constant with draws from the income distribution in this baseline specification. The coefficient of interactions of prices with both types of draws are insignificant. The interaction of the constant with income is just significant at the 10%-level. The positive coefficient implies that consumers with higher income generally have a higher valuation of the outside option.

Trends in markups resulting from the full random coefficient model look very similar to those of the nested logit model.

26. Using quasi-random sampling techniques like MLHS or Halton draws tends to improve precision of estimates. See Train (2009) for detailed explanations and simulations.

F Supplementary results

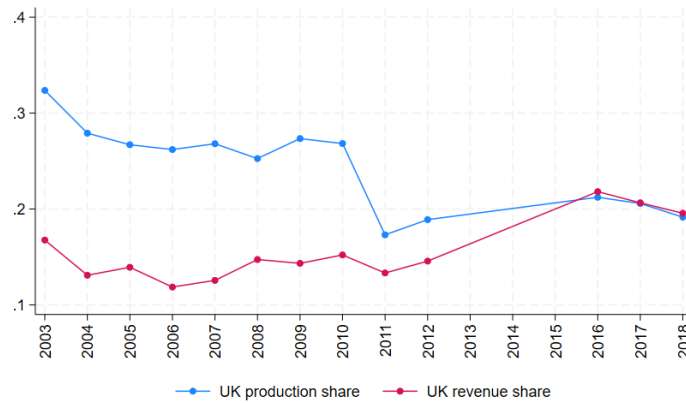
Table F.1: Demand Nest Summary Statistics

	(1)	(2)	(3)	(4)
	Nest 1	Nest 2	Nest 3	Nest 4
	mean/sd	mean/sd	mean/sd	mean/sd
Kerbside weight kg (average across trims)	1170.9 (178.3)	1593.3 (193.4)	1820.3 (257.3)	2101.8 (268.0)
HP (average across trims)	101.9 (26.4)	158.3 (41.4)	413.7 (114.2)	252.2 (78.5)
Price ('000£)	17.3 (5.5)	30.3 (8.0)	93.6 (56.5)	54.6 (17.4)
Observations	2562	2545	252	750

Note: table shows means (and standard deviations in parentheses) of weight, horsepower (HP), and prices by nest. We generate four nests using a k-clustering algorithm based on weight and horsepower.

For 2003-2012 and 2016-2018 we can use SMMT data on production and exports to compare our estimates of UK revenue shares with UK production shares. This comparison suggests our UK revenue share estimates are plausible, although the two are less comparable in earlier years, which could be due to differences in the SMMT methodology prior to 2016.

Figure F.1: UK revenue and production shares over time



Note: figure shows mean UK revenue and production shares. The former is calculated by inferring UK revenue from JATO data and taking total revenue from HO. The latter is calculated by comparing SMMT export and production data.